



An Integrated Force Myography and SVM-Based Machine Learning System for Enhanced Muscle Exertion Assessment in Industrial Settings

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Abstract

This study proposes a novel approach for objective muscle exertion assessment in industrial settings, combining force myography (FMG) and support vector machines (SVM), bridging the gap between subjective and objective assessments while addressing limitations of existing technologies. To improve FMG data quality, an in-house-built conditioning interface for force-sensing resistor (FSR) sensor was developed, enhancing sensitivity and reducing drift. A novel hybrid feature selection method, combining multicollinearity analysis with forward selection, was implemented to analyze 31 time and frequency domain features extracted from the data. Various kernel functions (linear, polynomial, Gaussian) for the SVM model were evaluated using k-fold cross-validation to identify the optimal configuration. The results revealed 6 key features for optimal classification of Ratings of Perceived Exertion (RPE) using SVM. The Fine Gaussian kernel SVM model achieved the highest performance, demonstrating 97.09% accuracy, 96.83% precision, 96.79% recall, and a 96.81% F1-score. However, the potential of the Medium Gaussian kernel with larger datasets merits further investigation. Despite limitations such as a small sample size and focus on a single muscle group, our findings establish a foundation for a systematic approach to objective muscle exertion assessment. Future research should address these limitations to improve generalizability and develop robust ergonomic assessments for real-world industrial applications.

Keywords Muscle contraction assessment · Force myography · Ratings of perceived exertion · Feature selection · Support vector machine

Abbreviations

FDS Flexor Digitorum Superficialis
FMG Force Myography
FSR Force Sensing Resistor

RPE Ratings of Perceived Exertion
sEMG Surface Electromyography
SVM Support Vector Machine

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1 Introduction

In industrial environments, optimizing worker safety and productivity is critical and hinges on the accurate assessment of muscle effort. Traditional methods for evaluating muscle exertion, often employed by ergonomists and physical therapists, primarily involve subjective assessments of voluntary muscle contractions [1]. However, these methods are prone to individual variability, biases, and recall errors, leading to inaccuracies in estimating actual muscle effort [2–5]. This discrepancy undermines the effectiveness of interventions aimed at preventing musculoskeletal disorders (MSDs).

Recent advances have focused on objective, non-invasive technologies. Surface Electromyography (sEMG) is rec-



ognized as the gold standard for non-invasive monitoring of muscle activity through wearable sensors [6]. However, its industrial application is hindered by various factors including electrode and skin properties, electrode-skin interference and electromagnetic noise [7]. Mechanomyography (MMG), on the other hand, records mechanical vibrations produced by contracting muscle fibers utilizing microphones or accelerometers provides a solution less susceptible to electrical noise but struggles with dynamic movements [8–10].

Force Myography (FMG) represents a significant advancement in this field. FMG measures the muscle force generated at the surface through changes in muscle volume, providing an objective and quantitative analysis while overcoming the limitations of sEMG and MMG [11, 12]. Based on force-sensing resistor (FSR) technology, FMG's insensitivity to electrical noise and its applicability in both static and dynamic conditions make it a suitable alternative for industrial applications [13, 14]. Moreover, current evidence suggests FMG can effectively classify muscle contraction levels, demonstrating the potential for machine learning applications in monitoring muscle function [13, 15–17].

Despite its potential, FMG faces challenges such as variations in subject anthropometry, sensor sensitivity, and measurement drift, which can affect signal acquisition and the accuracy of machine learning models [11, 13, 18]. Recent evidence suggests that conditioned FMG signals, combined with advanced machine learning techniques like support vector machine (SVM), can improve muscle contraction assessments [15, 19]. However, the efficacy of SVM heavily depends on the selection of an appropriate kernel function [20].

Therefore, this study aims to enhance FSR sensitivity and address machine learning challenges by developing a novel data-driven approach for muscle contraction assessment. This includes optimizing feature selection and evaluating different SVM kernel functions to improve the prediction of muscle Ratings of Perceived Exertion (RPE). By advancing FMG technology and machine learning applications, this research seeks to bridge the gap between objective muscle effort measurements and workers' subjective experiences in industrial settings.

2 Materials and Methods

2.1 Materials

The muscle contraction results in changes in muscle volume, which can be non-invasively measured using the FMG technique. FMG is commonly employed to monitor functional movements and hand gestures by observing volumetric alterations in limbs. These changes correlate with radial forces generated by the muscles, which can be quantified using FSR.

In this study, we used an Interlink-402 FSR (See Fig. 1a), featuring two interdigitated electrodes atop a semi-conductive polymer layer in a shunt mode configuration (See Fig. 1, Section "Force Sensing Resistors: An Overview of the Technology", p. 5 in [21]).

To mitigate the limitations of FSR performance in tracking muscle contractions and ensure reliable measurements, an in-house built conditioning electrical and mechanical interfaces were developed. The mechanical interface included a rigid dome and a rigid back to enhance sensor-to-skin contact and prevent sensor bending (See Fig. 1c), following recommendations of Esposito et al. [11]. Additionally, the electrical interface was integrated to increase the sensitivity and reduce drift. Although the manufacturer, Interlink Electronics, suggests using a simple voltage divider circuit (Refer to Figure 9 in the section "Suggested Electrical Interfaces: Basic FSRs", p. 18 in [21]), Xiao and Menon indicate that this configuration diminishes sensor sensitivity and amplifies drift [13]. Consequently, the input voltage was adjusted to a constant 5V. Furthermore, a current-to-voltage converter was implemented to ensure response linearization, as recommended by the manufacturer (Refer to Figure 15 in the section "FSR Current-to-Voltage Converter", p. 23 in [21]). For precise control of the output gain of the FSR configuration, an operational amplifier (op-amp) and a resistor were utilized. This study used an LM358P op-amp and a 1KOhm resistor to create a constant voltage across the sensor and an output voltage proportional to the current flowing through it (See Fig. 1b). To ensure optimal sensor-to-skin contact during muscle contraction, the FMG system was mounted on an elastic strap that could be wrapped around the limb.

2.2 Methods

2.2.1 Experimental Protocol

Twelve healthy, able-bodied subjects (six females and six males, aged 21–26 years; height: 163–180 cm, weight: 46–75 kg) were recruited for the experiment. All subjects had no musculoskeletal conditions, chronic diseases, or history of muscle or bone surgeries, and were not taking any medications or substances that could potentially impact their physical performance. The conditioned FSR-sensor, operating at a 1 kHz sampling rate, was placed on the belly of Flexor Digitorum Superficialis (FDS) muscle in the dominant upper limb to assess the muscle's activity. This muscle exhibits the highest activation during power grip exercises [22]. In this study, RPE were defined in reference to the Borg Category-Ratio CR10 scale [23]. This scale is a widely utilized tool for assessing physical activity based on RPE. A score of 0–2 indicates a very light RPE, a score of 3–4 indicates a light RPE, a score of 5–6 indicates a moderate RPE, a score of

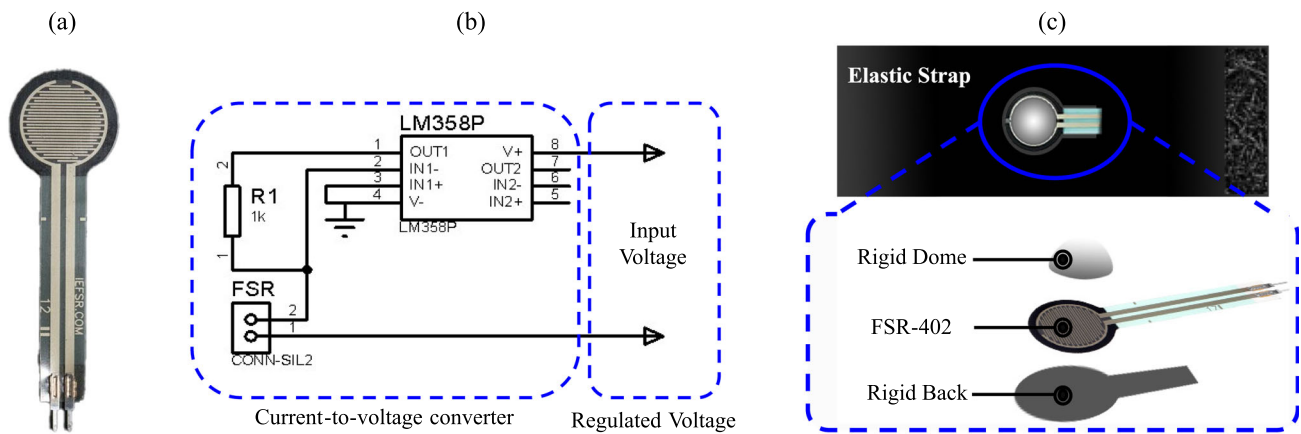


Fig. 1 Illustration of the FMG signal acquisition system: **a** FSR-Interlink 402 sensor, **b** electrical interface for FSR sensor conditioning, and **c** mechanical interface for muscle force measurement

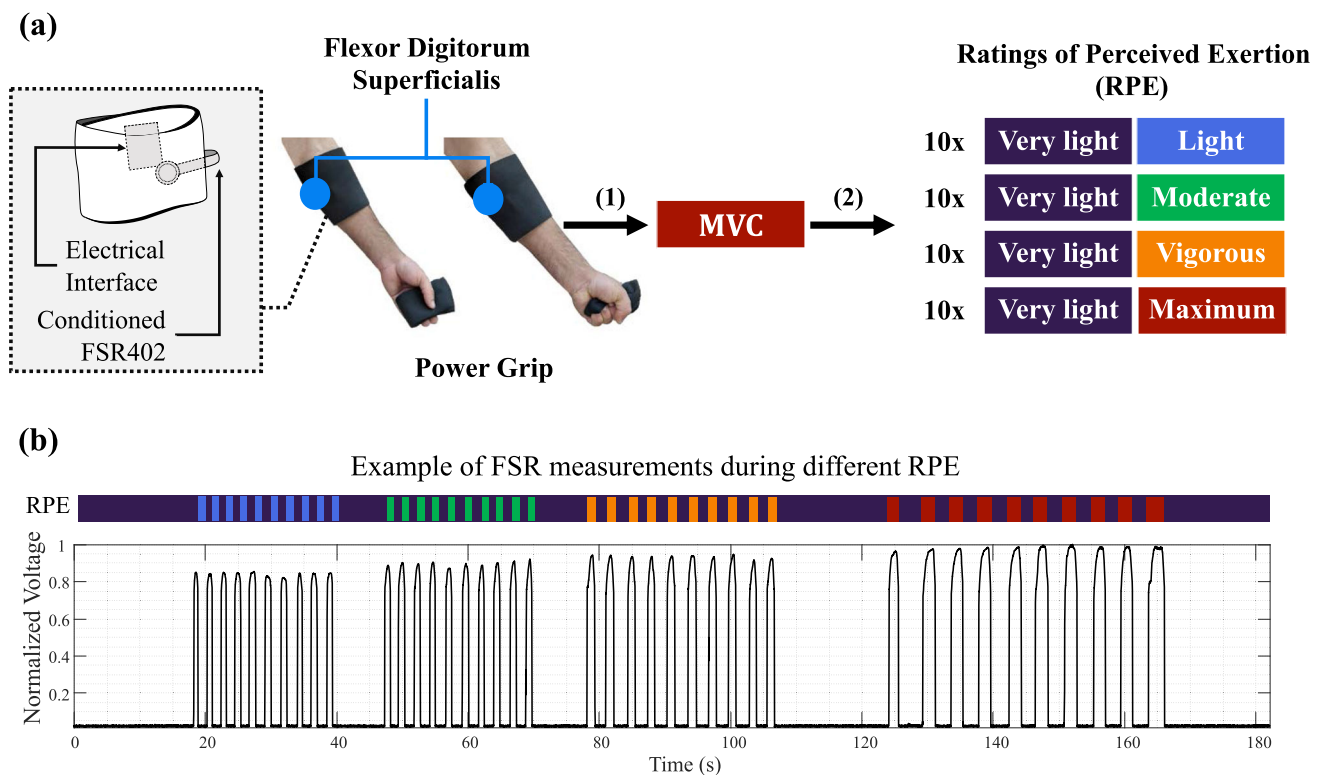


Fig. 2 Comprehensive illustration of **a** the experimental protocol for recordings of muscle RPE using FMG, along with **b** FMG signal sample

7–8 indicates a vigorous RPE, and a score of 9–10 indicates a maximum RPE [23].

Participants began each session by performing a maximum voluntary contraction, followed by 10 repetitions of hand power grip at each level of exertion, maintaining a self-selected, comfortable pace to prevent fatigue (See Fig. 2a). Prior to data collection, each participant received a thorough explanation of the Borg CR10 scale. Additionally, they were given the opportunity to familiarize themselves with the equipment and minimize the potential for inaccuracies in

rating exertion levels by practicing before the experimental trials. In total, 46 experiments were carried out, with Fig. 2b showcasing a sample of the recorded signals.

In the preprocessing stage, the signal was screened to remove any data that may have been affected by sensor failure or poor contact. Only 40 tests with most consistent data were included in this study. No additional digital signal processing techniques, such as filtering, were employed prior to feature extraction. Following this, all consistent recorded signals were normalized based on the maximal voluntary



contraction measured with FSR. This normalization process accounted for individual variations in RPE, enabling comparisons between subjects. A 15% threshold of the maximal value was utilized to identify activity zones within the signals. This threshold facilitated the detection of activity segments in undisturbed and stable signals, as recommended by Staude and Wolf [24]. Subsequently, the activity regions were categorized into five different RPE, as shown in Fig. 2b.

2.2.2 Feature Extraction

For feature extraction, a non-overlapping 200 ms window was used, representing the minimum interval between distinct muscle contractions [25]. With a sampling rate of 1 kHz, over 40 sessions each lasting approximately 3.2 min, a total of 7680000 data points was collected (i.e. 192000 data points per session). Using a time window of 200 ms, we obtained a total of 38400 observations. Thirty-one features, adopted from sEMG, were extracted from these observations. Specifically, twenty-one time-domain features and ten frequency-domain features were extracted, each chosen for their relevance and efficacy in the context of sEMG analysis.

In the time domain (refer to Table 3 in Appendix A), the Integrated FMG (IFMG) represents an overall estimate of the signal's absolute values [26, 27], reflecting the mean activity level. The mean absolute value (MAV) offers a similar perspective but is normalized for signal length, making it more consistent across different segment durations [28–30]. The Modified Mean Absolute Values (MAV1 and MAV2) introduce weighting strategies to emphasize specific parts of the signal, with MAV1 focusing on the central part and MAV2 accentuating the signal's edges [31]. The Simple Square Integral (SSI) measures the total power within a signal segment [30], and the Variance of FMG (VAR) quantifies the variability around the mean [26, 28, 30]. Root Mean Square (RMS) provides insight into the fluctuation magnitude [27, 29, 30], whereas the Log Detector (LOG) emphasizes larger amplitudes and is sensitive to outliers [28, 30]. Waveform Length (WL) and Average Amplitude Change (AAC) both capture the rate of change within the signal, but AAC normalizes this change over segment length [28–30]. The Difference Absolute Standard Deviation Value (DASDV) extends this concept by focusing on the variability of these changes [27, 30]. Willison Amplitude (WAMP) and Mean Absolute Value Slope (MAVS) [26, 28, 31] are used to detect muscle contractions and trends in muscle activity, respectively. The Absolute value of the Summation of the expth root (ASM) and the Absolute value of the Summation of Square root (ASS) [31, 32] offer alternative ways to analyze the signal intensity, while Cardinality (CARD) and Difference Absolute Mean Value (DAMV) [27, 33] provide insights into the signal's complexity and rate of change, respectively. The Average Energy (ME) quantifies the overall energy content

of the signal [34], and the Skewness (SKEW) and Standard Deviation (SD) [29, 30, 34–36] provide measures of distribution asymmetry and dispersion around the mean. Finally, the Coefficient of Variation (COV) offers a scale-independent measure of variability [37].

In the frequency domain (refer to Table 4 in Appendix A), Mean Frequency (MNF) and Peak Frequency (PFK) [31] characterize the dominant frequency content of the FMG signal, with MNF providing an average [38] and PFK identifying the most dominant frequency [31]. Mean Power (MNP) and Total Power (TTP) measure the overall energy distribution and total energy content in the frequency domain [31]. Median Frequency (MDF) divides the power spectrum into two equal parts [38], while the 1st, 2nd, and 3rd Spectral Moments (SM1, SM2, SM3) offer detailed insights into the mean, variance, and skewness of the frequency distribution [31]. The Power Spectrum Ratio (PSR) and Variance of central frequency (VCF) further analyze the localized frequency content and stability of muscle contractions [31]. These features, when integrated with time-domain attributes, provide a comprehensive and complementary perspective, offering insights into muscle activity that are not readily apparent in the time-based representation of the signal.

2.2.3 Feature Selection

This study introduces a hybrid approach designed to mitigate overfitting and enhance the stability of the model. This method combines multicollinearity analysis and forward feature selection (as shown in Fig. 3) to identify features that are not only predictive but also representative and explainable.

The initial phase involves conducting a multicollinearity analysis to assess the correlations between features. To conduct multicollinearity analysis, two primary metrics are employed: the Variance Inflation Factor (VIF) and Tolerance. These metrics are calculated for each predictor variable to assess the degree of multicollinearity. The Variance Inflation Factor (VIF) for the i^{th} predictor is calculated using Eq. 1, where R_i^2 is the coefficient of determination obtained by regressing the i^{th} predictor on all other predictors.

$$\text{VIF}_i = \frac{1}{1 - R_i^2} \quad (1)$$

Tolerance, the inverse of VIF, for the i^{th} feature is given by Eq. 2. It measures the proportion of variance of the i^{th} predictor not explained by the other predictors. Lower Tolerance values suggest higher multicollinearity, as more of the predictor's variance is shared with other predictors.

$$\text{Tolerance}_i = 1 - R_i^2 \quad (2)$$

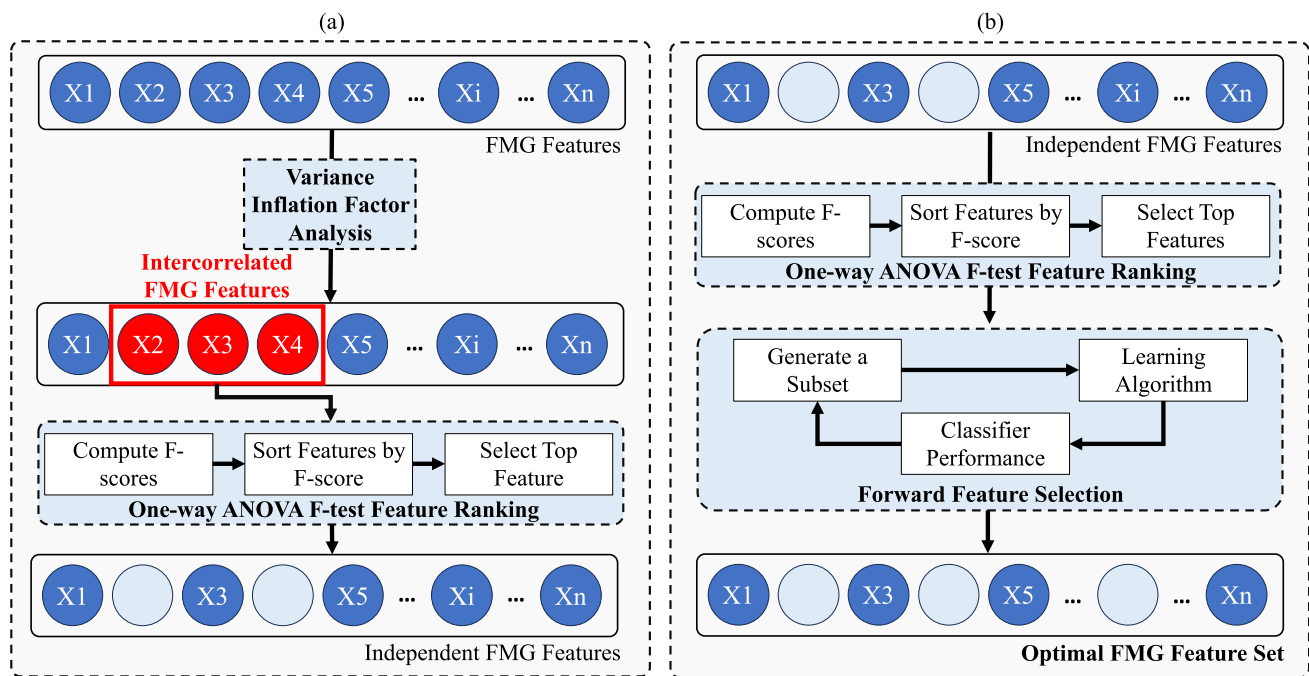


Fig. 3 Illustration of the Feature Selection and Reduction Methodology: A Hybrid Approach Combining **a** Multicollinearity Analysis and **b** Forward Feature Selection, with One-Way ANOVA Assessment of Feature Importance

In this study, features with VIF values above 10 or Tolerance below 0.1 are considered inter-correlated. These thresholds align with commonly accepted standards in statistical analysis for balancing feature retention and multicollinearity control [39]. If multicollinearity exists, the analysis ensures that only the most predictive features are retained, enhancing model reliability and stability [40].

The significance of individual features on the response variable is determined using one-way ANOVA F-statistics [41]. The ANOVA F-statistics test is used in feature ranking to compare models or sets of features based on their explained variance relative to the variance within the individual features. In ANOVA, the F-statistic follows the theoretical F-distribution under the null hypothesis. The null hypothesis assumes that all group means are equal. If the F-statistic is sufficiently large, it provides evidence against the null hypothesis, indicating that the group mean is different. In the context of feature selection, this indicates that a feature with a significant F-statistic enhances the model's predictive capability, with the highest F-statistic denoting the most predictive feature.

Building upon this foundation, Forward Feature Selection is subsequently employed to ensure that most informative features are included in the final subset. This wrapper method incrementally constructs a feature subset, starting with no features and incrementally adding the most significant feature from the original set in each iteration. This is based on its contribution to the model's performance, using the one-way

ANOVA F-statistics ranking of features. This iterative process continues until the addition of new features no longer significantly improves the model's performance [42].

2.2.4 Classifiers Performance Evaluation

To assess the performance of the SVM classifier to predict muscle RPE based on FMG signals, linear, polynomial, and gaussian kernels were tested. Linear kernel functions are typically used in cases where the data points are expected to be linearly separable [43]. This implies that in the context of muscle exertion data, the linear kernel function assumes the existence of a linear separation in the feature space, allowing classes of muscle exertion to be separated by a straight line or hyperplane. The linear kernel function is represented by Eq. 3, where x and y are input data points in the original feature space [43].

$$K(x, y) = x^T * y \quad (3)$$

Polynomial kernel functions (PF) are employed when the data points are expected to be nonlinear. Mathematically, the Polynomial kernel functions are represented by Eq. 4, where x and y are input data points in the original feature space, p is the polynomial order of the kernel function, and c is a user-defined constant [44].

$$K(x, y) = (x^T * y + c)^p \quad (4)$$



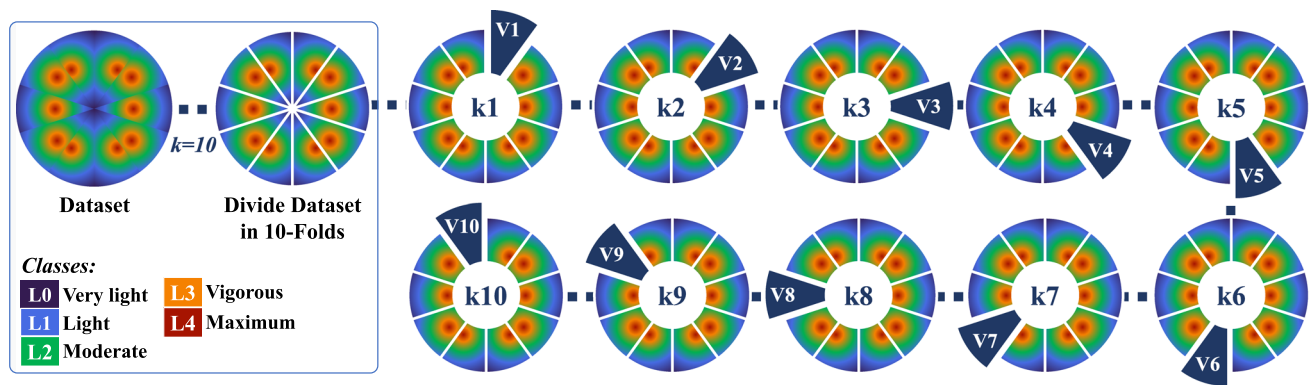


Fig. 4 Visual representation of the 10-fold cross-validation method used for model evaluation and validation: V_i ($i = 1..10$) represents the i th validation fold, while the remaining folds are dedicated to training

Gaussian kernel functions are typically used when the data points have complex, non-linear relationships. Mathematically, they are represented by Eq. 5, where x and y are input data points in the original feature space, and γ (gamma) is a user-defined hyperparameter [44]. The hyperparameter γ dictates the impact of each data point and the sensitivity of the kernel to the distance between data points in the feature space. A smaller γ leads to a wider Gaussian curve, allowing distant data points to contribute more to the similarity measure and resulting in a smoother decision boundary. On the other hand, a larger γ creates a narrower Gaussian curve, making the kernel more sensitive to distances between data points, which could lead to a more complex and tightly fitted decision boundary.

$$K(x, y) = \exp(-\gamma * \|x - y\|^2) \quad (5)$$

In this study, linear, quadratic (PF, $p = 2$), cubic (PF, $p = 3$), Fine Gaussian ($\gamma = \sqrt{p/4}$), Medium Gaussian ($\gamma = \sqrt{p}$), and Coarse Gaussian ($\gamma = 4\sqrt{p}$) kernel functions were investigated. To validate the performance of these SVM models, k-fold cross-validation was performed (Fig. 4). This technique entails dividing the data into k subsets, training the model on $k-1$ subsets, and evaluating the performance of the remaining subset. This process is repeated k times, with each subset being used once for an evaluation.

First, to provide balance between computational efficiency and accuracy, the optimal value of k for the k-fold cross-validation process was determined by evaluating SVM models with varying values of k , ranging from 4 to 20. The k value with the highest overall accuracy was selected as the optimal value for our study. Second, comparison between the kernel functions is performed using accuracy, precision, recall, and F1-score metrics. The accuracy metric indicates the proportion of correct predictions made by the model, calculated as the ratio of correctly classified instances to the total number of instances in the dataset. Precision measures

the correctness of positive predictions made by the classifier relative to the total number of positive predictions. Recall, or sensitivity, gauges the extent to which actual positive predictions are correctly identified. The F1-score offers insight into the balance between precision and recall, enabling comparison of different classifiers' performance. The closer these metrics are to 1, the better the result. These metrics can be calculated using Eqs. 6–9, in which True Positives (TP) represent correctly predicted instances, False Positives (FP) denote incorrectly predicted instances, True Negatives (TN) correspond to instances correctly predicted as negatives, and False Negatives (FN) stand for instances incorrectly predicted as negatives. These metrics offer valuable insights into the effectiveness of different kernel functions for rating muscle contraction levels.

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (6)$$

$$\text{Precision} = \frac{TP}{(TP + FP)} \quad (7)$$

$$\text{Recall} = \frac{TP}{(TP + FN)} \quad (8)$$

$$F1 - \text{score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{(\text{Precision} + \text{Recall})} \quad (9)$$

3 Results

3.1 Feature Selection

The multicollinearity analysis, which focuses on time-domain and frequency-domain features for assessing RPE, is outlined in Table 1. The analysis revealed significant inter-correlations among 22 features, including IFMG, MAV, MAV1, MAV2, SSI, VAR, RMS, LOG, WL, AAC, ASM, ASS, ME, DAMV, SD, MNF, PKF, MNP, TTP, MDF, SM1, and VCF (p - value < 0.05). This outcome is supported

Table 1 Results of multicollinearity analysis on time-domain and frequency-domain features for muscle REP assessment: examining predictor influence (R_i^2), variance inflation (VIF), predictive power (Tolerance), and statistical significance (p -value)

Predictor	R_i^2	VIF	Tolerance	p -value
IFMG	0.988	89.534	0.011	< 0.001
MAV	0.989	90.847	0.011	< 0.001
MAV1	0.988	90.344	0.011	< 0.001
MAV2	0.986	75.221	0.013	0.012
SSI	0.988	89.611	0.011	< 0.001
VAR	0.989	90.863	0.011	< 0.001
RMS	0.988	90.650	0.011	< 0.001
LOG	0.989	90.039	0.011	< 0.001
WL	0.948	19.471	0.051	1.054e−3
AAC	0.988	84.765	0.012	< 0.001
DASDV	0.886	8.823	0.113	< 0.001
WAMP	0.881	5.462	0.183	< 0.001
MAVS	0.388	1.636	0.611	< 0.001
ASM	0.988	89.168	0.011	< 0.001
ASS	0.981	52.788	0.019	< 0.001
ME	0.988	90.866	0.011	< 0.001
CARD	0.981	6.723	0.149	6.591e−3
DAMV	0.989	84.486	0.012	1.199e−2
SKEW	0.851	2.431	0.411	8.399e−3
COV	0.988	5.772	0.173	0.014
SD	0.588	36.795	0.027	< 0.001
MNF	0.826	90.908	0.011	0.016
PKF	0.972	26.760	0.037	< 0.001
MNP	0.988	88.812	0.011	< 0.001
TTP	0.962	88.794	0.011	0.012
MDF	0.988	90.908	0.011	0.006
SM1	0.963	27.352	0.037	< 0.001
SM2	0.883	8.558	0.117	< 0.001
SM3	0.799	4.983	0.201	< 0.001
PSR	0.874	7.992	0.125	0.038
VCF	0.960	25.280	0.040	< 0.001

by high R_i^2 values for these features ($R_i^2 > 0.7$). In contrast, DASDV, WAMP, MAVS, CARD, SKEW, COV, SM2, SM3, and PSR exhibited no significant inter-correlations, revealed by VIF values below 10 and tolerance values above 0.1 (p - value < 0.05).

To select the most predictive feature among the 22 multicollinear ones, the one-way ANOVA feature ranking is shown in Fig. 5a, in which the ranks are normalized by the maximal F-statistic reported in the feature ranking. The result suggests that ASM was the highest-ranked feature. Consequently, only this feature will be added to previous selected features with low levels of multicollinearity. In sum, based on the multicollinearity analysis results, 10 features (ASM, DASDV,

WAMP, MAVS, CARD, SKEW, COV, SM2, SM3, and PSR) are deemed suitable for RPE classification.

Figure 5b displays the result of the one-way ANOVA feature ranking of the 10 selected independent features. Figure 6 shows the accuracy of models in RPE in function of the number of independent features added incrementally, beginning with the feature of the highest rank.

The results reveal that accuracy does not increase as the number of features increases. Specifically, using more than six features did not improve classification accuracy for linear, quadratic, and gaussian SVM classifiers. However, for the cubic SVM classifier, including additional features beyond the sixth led to reduced accuracy, likely due to increased computational complexity and information redundancy. To mitigate this issue, the first 6 important features (ASM, COV, SKEW, SM2, SM3, and WAMP) were selected for the classifiers performance comparisons.

3.2 Classifiers Performance Evaluation

Figure 7 presents the performance of SVM models with varying k values, ranging from 4 to 20. The highest overall accuracy of 94.22% was achieved with k equal to 10, which provided insights for optimizing the model by selecting the best k value for the cross-validation process.

The performance of the chosen kernel functions in classifying muscle RPE was evaluated in terms of accuracy, precision, recall, F1- score and Training Time, as shown in Table 2. Additionally, the performance of each kernel function in classifying the different muscle RPE is presented in Fig. 8.

The overall accuracy, precision, recall, and F1-score calculated for all classifiers exceed the designed threshold of 70%, confirming the suitability of SVM classifiers for FMG classification applications. The Fine Gaussian kernel achieves the highest overall accuracy (97.09%), precision (96.83%), recall (96.79%), and F1-score (96.81%). Conversely, the cubic SVM classifier yields the lowest overall accuracy (80.26%), precision (78.45%), recall (80.28%), and F1-score (79.36%). Upon closer examination, the cubic SVM classifier's performance raises concerns, as illustrated in Fig. 8. The recall values for the "Vigorous" and "Maximum" classes are as low as 49.1% and 69.5%, respectively, suggesting significant misclassification or failure to recognize instances in these classes, with misclassification rates of 50.9% and 30.5%, respectively. Additionally, the cubic SVM presents poor performances in classifying the "Vigorous" class, as indicated by its low precision value of 25.9%, signifying that a considerable portion (74.1%) of instances classified as "Vigorous" were falsely classified. The cubic SVM's poor performance is further underscored by its F1-score value of 30.28%, which emphasizes the model's



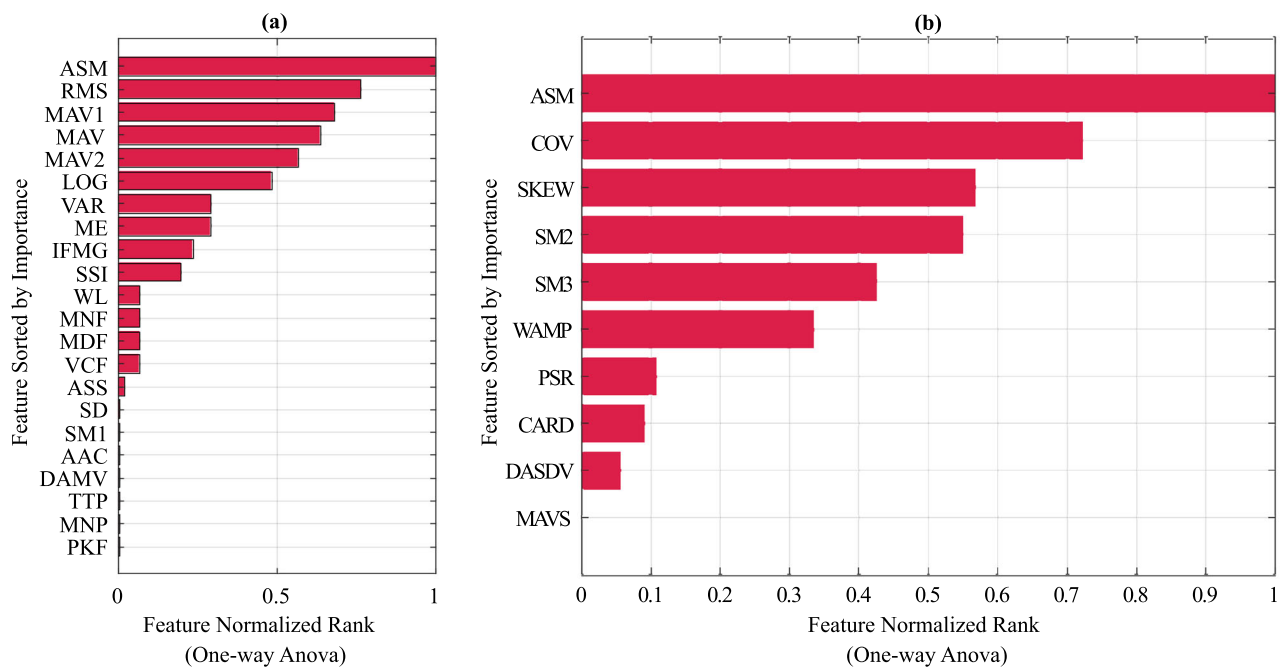


Fig. 5 Results of One-way ANOVA Feature Ranking of Features **a** with Evidence of Multicollinearity, and **b** with No Evidence of High Multicollinearity

Fig. 6 Impact of the number of features on the accuracy of SVM classifiers in FMG-based muscle RPE

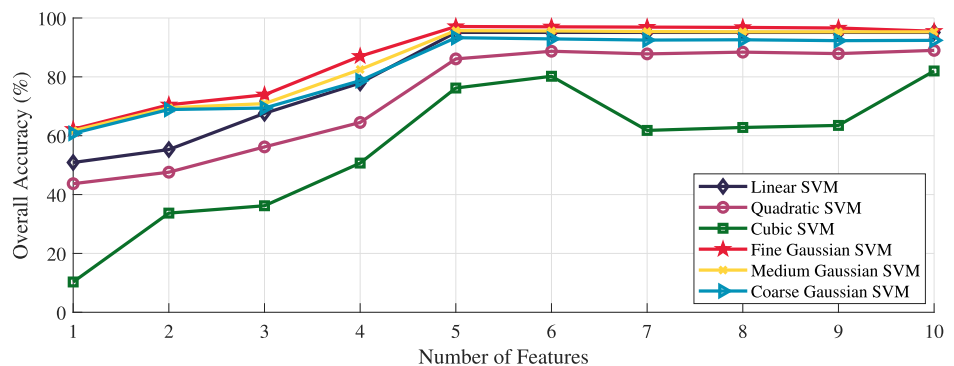
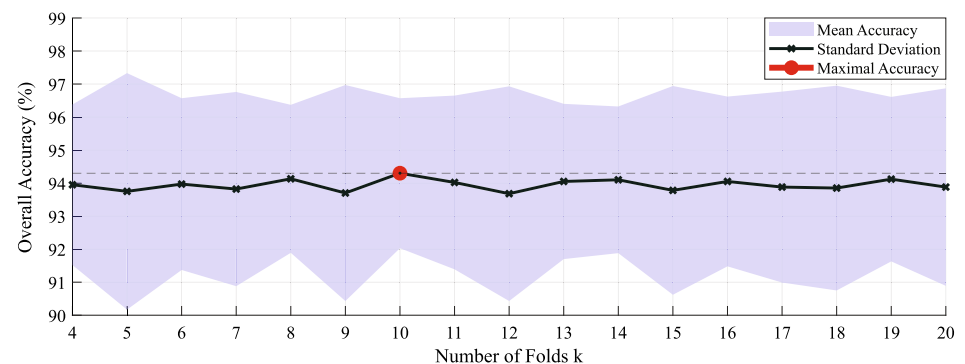


Fig. 7 Impact of the number of folds used in k-fold cross-validation on the overall accuracy of SVM classifiers in FMG-based muscle RPE



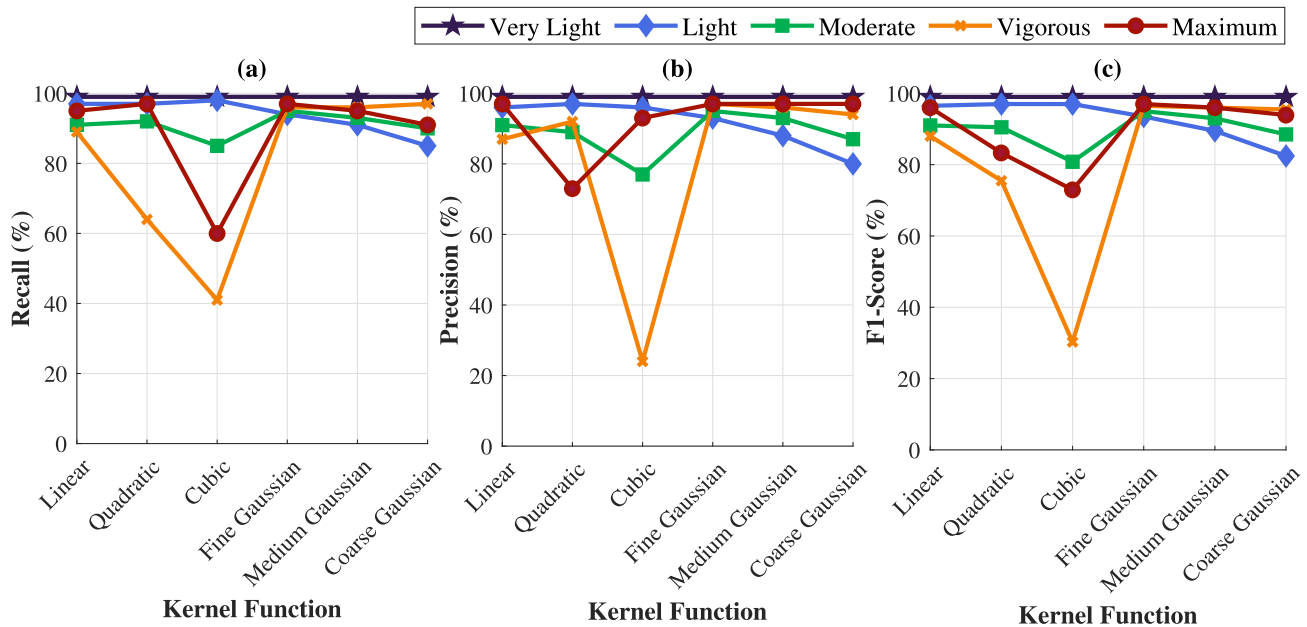
imbalance between precision and recall. In contrast, the Fine Gaussian kernel function shows high values of precision, recall, and F1-score in classifying the different classes (between 95.5% and 99%).

4 Discussion

This study aimed to bridge the gap between quantitative muscle activity assessment and the subjective RPE, reflect-

Table 2 Overall SVM accuracy, precision, recall, F1-Score, and training time

Metric	Kernel function					
	Linear	Quadratic	Cubic	Fine G	Medium G	Coarse G
Accuracy (%)	95.14	88.74	80.26	97.09	95.60	92.96
Precision (%)	94.51	90.66	78.45	96.83	95.11	91.86
Recall (%)	94.73	90.49	80.28	96.79	95.30	92.95
F1-score (%)	94.62	90.58	79.36	96.81	95.21	92.40
Training Time (s)	635.5	3172.3	2585.9	176.53	129.3	154.79

**Fig. 8** Recall, precision, and F1-score by class using linear, quadratic, cubic, fine gaussian, medium gaussian, and coarse gaussian SVM

ing the physical demands of tasks in industrial settings. We investigated the relationship between FMG signals, measured during power grip tasks, and RPE ratings reported by participants. A hybrid feature selection methodology was implemented to identify the most informative features from the FMG data, and a SVM classification approach was used to establish the link between the chosen features and the reported exertion levels.

A distinctive set of 6 informative FMG features (ASM, COV, SKEW, SM2, SM3, WAMP) was identified for RPE classification. The selected features are relevant for characterizing muscle effort and are easily interpretable, aligning with the need for clear physiological understanding. For instance, ASM offers insights into overall signal power, reflecting a comprehensive view of the FDS muscle's response during the power grip task [32]. COV, on the other hand, captures relative variability within the FMG signal, potentially distinguishing between steady and variable exertion patterns [37]. Furthermore, SKEW sheds light on potential muscle asymmetries that might arise during exer-

tions [45]. Spectral moments, SM2 and SM3, contribute to characterizing muscle exertion levels by assessing power distribution within the FMG signal [46]. Finally, WAMP helps gauge muscle activation level, potentially serving as an indicator of muscle fatigue through detection of rapid contractions and relaxations [26]. By combining multicollinearity analysis with forward selection, we ensured that the selected features were informative, minimally correlated, and directly relevant to physiological processes underlying RPE. Unlike automatic selection techniques, which depend on less relevant criteria like individual feature variance, our manual approach prioritizes explainable features. This approach, while time-consuming, confirms previous research highlighting the benefits of explainable features for enhancing model accuracy and facilitating the development of evidence-based strategies to manage physical demands in practical settings [47–51].

Our study further emphasizes the effectiveness of the Gaussian kernel SVM for classifying RPE based on FMG signals. This finding corroborates previous research find-

ings demonstrating the efficiency of Gaussian SVM in tasks involving muscle activity patterns. Using FMG for several applications, such as motion recognition [30], hand gesture classification [52], and estimating isometric torque generated by the wrist and forearm [53]. The success of the Gaussian kernel in our study can likely be attributed to its ability to effectively model the nonlinear relationships between the FMG features and RPE ratings. This characteristic makes Gaussian kernels more flexible and suitable for complex data compared to linear or polynomial kernels, particularly advantageous for modeling the intricate dynamics of muscle activity during power grips [54–57]. The Fine Gaussian kernel specifically achieves a balance between efficient computation and high classification accuracy. This is because it has a smaller kernel scale ($\gamma = \sqrt{p/4}$) compared to the medium ($\gamma = \sqrt{p}$) and coarse ($\gamma = 4\sqrt{p}$) Gaussian functions. This allows it to more effectively capture subtle variations within the FMG data, aligning with prior research which has shown that the performance and accuracy of SVM, can significantly depend on the specific application and experimental setup [58–61]. Regardless, the Fine Gaussian SVM exhibited higher computational demands compared to the Medium Gaussian SVM. This raises the question of whether a larger dataset could improve the Medium Gaussian SVM's performance. However, the relationship between dataset size and model efficiency in SVM classification is complex [62, 63]. Further investigation is required to evaluate the Medium Gaussian SVM performance with a larger dataset.

While this study lays the groundwork for understanding muscle RPE in grip tasks and identifying FMG as a potential tool for real-time monitoring, we must acknowledge its limitations when considering the complexities of industrial grip tasks. Among these limitations are the small sample size of only 12 young and healthy participants, the use of a single FSR sensor on the FDS muscle and the focus on a simple isometric power grip. These factors may limit the generalizability of the results to broader populations, varied grip types, and real-world industrial settings. Incorporating these limitations strengthens the potential impact of this research. By reducing model complexity without sacrificing predictive and interpretability power of selected features, the development of lightweight and accurate real-time muscle activity monitoring systems will be advanced. These systems could be invaluable for musculoskeletal disorders and injury prevention and rehabilitation, providing precise and personalized insights into muscle dynamics, ultimately helping to mitigate muscle strain and fatigue in complex industrial settings. Further research would aim to overcome these limitations by incorporating a larger set of participants, employing multi-sensor muscle monitoring, and exploring specific grip types encountered in real-world industrial settings. These advancements are critical for developing more robust ergonomic assessments and tool designs, ultimately

contributing to the reduction of muscle strain and fatigue in real industrial environments.

5 Conclusion

This study presented a novel approach for quantifying muscle contractions and predicting muscle RPE in industrial settings, overcoming limitations of traditional subjective methods. By harnessing the potential of FMG and integrating machine learning techniques, particularly SVM classifiers, we aimed to bridge the gap between objective muscle effort assessment and subjective worker experiences.

Our findings demonstrate the efficacy of FMG in providing an objective, quantitative analysis of muscle force, which is crucial for ensuring worker safety and optimizing productivity in industrial environments. The hybrid feature selection methodology combining multicollinearity analysis and forward feature selection identified key features that are most indicative of muscle exertion levels. By narrowing down the feature set to the most relevant ones, the SVM classifiers' performances were enhanced, reducing the risk of overfitting and improving their generalizability. The findings revealed that including more than 6 features did not improve the classifiers' accuracy. This insight is crucial for developing streamlined and computationally efficient models for real-time muscle exertion monitoring. Furthermore, the evaluation of different SVM kernel functions provided a clear understanding of their effectiveness in classifying muscle exertion levels. The Fine Gaussian kernel SVM stood out for its superior performance, achieving the highest overall accuracy, precision, recall, and F1-score. This kernel's ability to capture the nuances in the FMG data makes it particularly suited for this application.

While this study establishes the promise of FMG for real-time muscle activity monitoring in ergonomics and industrial safety, further research strengthens the evidence base and generalizability of these findings. Future studies should incorporate a more diverse larger participant pool and explore the application of FMG to various grip types encountered in real-world settings. Addressing these limitations will solidify the potential of FMG applications, ultimately contributing to improved worker safety and productivity in industrial environments.

Appendix A: Comprehensive Tables of FMG Signal Analysis Features

See Tables 3 and 4.

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Table 3 Summary of time-domain features used in the FMG signal analysis

Feature	Formula	Ref
Integrated FMG	$IFMG = \sum_{i=1}^N x_i $	[26, 27]
Mean absolute value	$MAV = \frac{1}{N} \sum_{i=1}^N x_i $	[28–30]
Modified mean absolute value type 1	$MAV1 = \frac{1}{N} \sum_{i=1}^N \omega_i x_i $	[31]
Modified mean absolute value type 2	$MAV2 = \frac{1}{N} \sum_{i=1}^N \omega_i x_i $	[31]
Simple square integral	$SSI = \sum_{i=1}^N x_i^2$	[30]
Variance of FMG	$VAR = \frac{1}{N-1} \sum_{i=1}^N x_i^2$	[26, 28, 30]
Root mean square	$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$	[27, 29, 30]
Log detector	$LOG = e^{\frac{1}{N} \sum_{i=1}^N \log x_i }$	[28, 30]
Waveform length	$WL = \sum_{i=1}^{N-1} x_{i+1} - x_i $	[28–30]
Average amplitude change	$AAC = \frac{1}{N} \sum_{i=1}^{N-1} x_{i+1} - x_i $	[30]
Difference absolute standard deviation value	$DASDV = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (x_{i+1} - x_i)^2}$	[27, 30]
Willison amplitude	$WAMP = \sum_{i=1}^{N-1} f(x_i - x_{i+1})$	[26, 28]
Mean absolute value slope	$MAVS = MAV_{k+1} - MAV_k$	[31]
Absolute value of the summation of the expth root	$ASM = \left \frac{\sum_{n=1}^N (x_n)^{\exp}}{N} \right $	[31]
Absolute value of the summation of square root	$ASS = \left \sum_{n=1}^N x_n^{\frac{1}{2}} \right $	[32]
Cardinality	$CARD = \sum_{i=1}^{N-1} y_i - y_{i+1} $	[33]
Difference absolute mean value	$DAMV = \frac{1}{N} \sum_{i=1}^{N-1} x_{i+1} - x_i $	[27]
Average energy	$ME = \frac{1}{N} \sum_{i=1}^N x_i^2$	[34]
Skewness	$SKEW = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \bar{x}}{\sigma} \right)^3$	[35, 36]
Standard deviation	$SD = \sqrt{\frac{n \sum_{i=1}^N x_i^2 - \left(\sum_{i=1}^N x_i \right)^2}{n(n-1)}}$	[29, 30, 34, 35]
Coefficient of variation	$COV = \frac{SD}{\frac{1}{N} \sum_{i=1}^N x_i}$	[37]

x_i ($i = 1 : N$) represent FMG segments where N is the total number of segments in FMG signal

Table 4 Summary of frequency domain features in FMG signal analysis

Feature	Formula	Ref
Mean frequency	$MNF : \left\{ P = \frac{\sum_{i=1}^N f_i P_i}{\sum_{i=1}^N P_i} \right\}$	[38]
Peak frequency	$PFK : \{ P = \max(P_i), \quad i = 1, \dots, N \}$	[31]
Mean power	$MNP = \frac{\sum_{i=1}^N P_i}{N}$	[31]
Total power	$TTP = \sum_{i=1}^N P_i$	[31]
Median frequency	$MDF : \left\{ P = \sum_{i=1}^{MDF} P_i = \sum_{i=MDF}^N P_i = \frac{1}{2} \sum_{i=1}^N P_i \right\}$	[38]
1st spectral moment	$SM1 = \sum_{i=1}^N P_i f_i$	[31]
2nd spectral moment	$SM2 = \sum_{i=1}^N P_i f_i^2$	[31]
3rd spectral moment	$SM3 = \sum_{i=1}^N P_i f_i^3$	[31]
Power spectrum ratio	$PSR = \frac{\sum_{i=f_0+n}^{f_0+n} P_i}{\sum_{i=-\infty}^{+\infty} P_i}$	[31]
Variance of central frequency	$VCF = \frac{SM2}{SM1} - \left(\frac{SM1}{SM0} \right)^2$	[31]

P_i is the FMG power spectrum at frequency bin i ($i = 1 : N$) where N is the total number of segments in FMG signal. f_i is its respective frequency



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Declarations

Conflict of interest The authors declare that they have no Conflict of interest to report regarding the present study

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